PROBLEMS Week: 26 March / 26 Maart 2018

All the indicated numbers refer to the "Problems" section in Serway; e.g. P1.9 is problem 9 from chapter 1.

Alle nommers verwys na die "Problems" afdeling in Serway; bv. P1.9 is probleem 9 uit hoofstuk 1.

OQ 5.1	OQ 5.2	OQ 5.5	OQ 5.7	OQ5.11
CC 5.1	CQ 5.2	CQ 5.4	CQ 5.5	
CQ 5.10	CQ 5.19			

P5.4
P5.19 (a) a) a = 5:00 m/s^2 at θ= 36.9° (b) a = 6.08 m/s^2 at 25.3°
P5.22 (a) 181° counter-clockwise from the positive x-axis (b) 11.2 kg (c) 37.5 m/s (d) $\vec{\mathbf{v}}_f = \left(-37.5\hat{\mathbf{i}} - 0.893\hat{\mathbf{j}}\right)$ m/s
P5.29 (a) 7.0 m/s^2 horizontally to the right (b) 21 N (c) 14 N horizontally to the right

P5.33 T_1 = 254 N, T_2 = 165 N, T_3 = 325 N
P5.34 Proof
P5.36 (a T_1 = 31.5 N, T_2 = 37.5 N, T_3 = 49.0 N, (b) T_1 = 113 N, T_2 = 56.6 N, T_3 = 98.0 N
P5.37 F_1 = 8.66 N east
P5.47 x_f = 3.73 m
P5.49 (a) 2.20 m/s^2 (b) 27.4 N
P5.60 (a) (b) θ = 55.2° (c) n = 167 N
P5.64 (a) FBD's (b) 2.31 m/s^2, down for m1, left for m2, and up for m3 (c) T_{left} = 30 N, T_{right} = 24.2 N (d) Explanation

P5.66 (a) P_{max} = 48.6 N and P_{min} = 31.7 N (b) Explanation (c) The block cannot slide up the wall. If P < 62.7 N, it will slide down the wall.

P5.88 [TO BE DONE DURING THE FOLLOWING WEEK'S TUTORIAL]

P5.88 Set the static friction coefficient $m_s = \mu_s$.
(a) $M = 3m\sin\theta$ (b) $T_1 = 2mg\sin\theta$, $T_2 = 3mg\sin\theta$ (c) $a = \frac{g\sin\theta}{1+2\sin\theta}$, $T_1 = 4mg\sin\theta\left(\frac{1+\sin\theta}{1+2\sin\theta}\right)$, $T_2 = 6mg\sin\theta\left(\frac{1+\sin\theta}{1+2\sin\theta}\right)$
(e) $M_{\max} = 3m(\sin\theta + \mu_s\cos\theta)$ (f) $M_{\min} = 3m(\sin\theta - \mu_s\cos\theta)$
(g) $T_{2,\max} - T_{2,\min} = M_{\max}g - M_{\min}g = 6\mu_s mg\cos\theta$
P5.89 (a) Proof (b) If $\tan\theta < \frac{1}{\mu_s}$ is not met, no value of P can move the crate.